

ANALYST REPORT

Climate Change Twitter Sentiment & Discourse Analysis

Public Opinion Trends Across 15.7 Million Tweets | 2006 – 2019

Field	Detail
TO:	Head of Growth & Head of Policy Research
FROM:	Balogun Bunmi, Data Analyst
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RE:	Climate Change Twitter Sentiment & Discourse Analysis 2006-2019
TOOL:	PostgreSQL (SQL) Power BI Desktop
DATASET:	The Climate Change Twitter Dataset — Mendeley Data v2
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1. Executive Summary

Climate change has become one of the most polarizing and consequential topics of public discourse in the digital age. This report presents a comprehensive analysis of 15,789,411 tweets spanning 14 years from 2006 to 2019, examining how global public sentiment, stance, and discourse patterns have evolved across topics, regions, and demographic groups. The analysis was conducted using PostgreSQL for data processing and Power BI for interactive visualization, drawing on the Climate Change Twitter Dataset published by Democritus University of Thrace on Mendeley Data.

The headline finding is that climate belief has grown dramatically over the 14-year period, from just 16.67% of climate tweets in 2006 to 79.40% by 2018, while climate denial has steadily declined from 8.33% to 6.18%. Despite this encouraging shift, the discourse remains deeply toxic: 28.67% of all climate tweets contain aggressive language, with deniers generating aggression at nearly double the rate of believers (42.65% vs 27.33%). North America is the epicenter of both climate denial and aggressive discourse, while Asia and Africa maintain the most constructive and positive climate conversations globally.

Metric	Value	Significance
Total Tweets Analysed	15,789,411	14 years of global climate discourse
Overall Avg Sentiment	-0.0047	Near-neutral — world deeply divided on climate
Total Believers	11,292,424 (71.52%)	Dominant stance — growing year on year
Total Deniers	1,191,386 (7.55%)	Minority but disproportionately aggressive
Aggressive Tweets	4,527,267 (28.67%)	Nearly 1 in 3 tweets contains hostile language
Geolocated Tweets	5,307,538 (33.61%)	Twitter location-sharing is opt-in by users
Dataset Coverage	2006 to 2019	14 years — 2020-2022 not in public release
Total Disaster Events	4,913	Across 203 countries and 9 disaster types
People Affected by Disasters	2.38 billion	Scale of real-world climate impact in same period

table 1: overall dataset summary statistics — all figures from PostgreSQL queries on the cleaned tweets_clean table

2. Data Understanding & Preparation

2.1 Dataset Overview

The Climate Change Twitter Dataset is the most comprehensive publicly available collection of climate-related tweets to date, published by Effrosynidis et al. at Democritus University of Thrace (DOI: 10.1016/j.eswa.2022.117541). The dataset contains 15,789,411 tweet records spanning from June 2006 to December 2019, each tagged with seven dimensions of information. A companion disasters dataset containing 4,913 climate-related disaster events across 203 countries was used for contextual correlation analysis. Since Twitter forbids making tweet text publicly available, the dataset contains metadata and analytical labels rather than raw tweet content. Both datasets were loaded into PostgreSQL via the `\copy` command to handle the volume safely.

Column	Data Type	Description	NULL Rate
created_at	TIMESTAMP	Date and time the tweet was posted (UTC)	0%
tweet_id	BIGINT	Unique identifier for each tweet	0%
lng	DOUBLE PRECISION	Longitude coordinate of tweet origin	66.4%
lat	DOUBLE PRECISION	Latitude coordinate of tweet origin	66.4%
topic	TEXT	Climate topic category — 10 distinct categories	0%
sentiment	DOUBLE PRECISION	Sentiment score from -1 (most negative) to +1 (most positive)	0%
stance	TEXT	Author climate stance: Believer, Denier, or Neutral	0%
gender	TEXT	Inferred author gender: Male, Female, or Undefined	0%
temperature_avg	DOUBLE PRECISION	Deviation from historical average temperature at tweet location	66.4%
aggressiveness	TEXT	Whether tweet contains aggressive language: aggressive or not aggressive	0%

table 2: data dictionary — climate change twitter dataset column definitions

2.2 Data Cleaning Decisions

Before any analysis was conducted, a systematic data quality audit was performed across all tables. The audit revealed that the dataset is remarkably clean for its scale. Zero duplicate tweet IDs were found, zero out-of-range sentiment scores were identified, and all coordinate values fell within valid geographical bounds. The primary data quality issue was the absence of geolocation data for 66.4% of tweets, a structural characteristic of the dataset reflecting Twitter's opt-in location sharing policy. Temperature deviation values were absent for the same 10,481,873 records since temperature is derived from location coordinates. A statistical check confirmed that temperature values fell within a realistic range of -23.29 to +21.00 degrees Celsius, with a mean of +1.25 and median of +1.21, confirming approximately normal distribution and the absence of extreme outliers.

Issue	Finding	Decision & Rationale
Missing coordinates (lng/lat)	10,481,873 records (66.4%) lack location data	Retained with has_geolocation flag — Twitter opt-in design means absence is expected, not erroneous
Topic name typo	Topic stored as 'Importance of Human Intervention'	Standardized to 'Importance of Human Intervention' using ILIKE in tweets_clean creation
Aggressiveness formatting	Values stored as lowercase 'aggressive' / 'not aggressive'	Standardized to Title Case using INITCAP(TRIM()) for consistency
Sentiment range validation	0 records found outside -1 to +1 range	No action required — all 15.7M sentiment values are valid
Duplicate tweet IDs	0 duplicates found across all 15,789,411 records	No action required — dataset is structurally clean
Temperature outliers	Range -23.29 to +21.00. Mean (+1.25) nearly identical to median (+1.21)	No action required — data follows normal distribution, no encoding errors like -999
Disaster NULL values	Some total_deaths and total_affected fields were NULL	Replaced with 0 using COALESCE — absence of record interpreted as no reported casualties

table 3: data cleaning decisions with rationale — applied in tweets_clean and disasters_clean table creation

3. Descriptive Analytics

3.1 Tweet Volume and Growth Trends

Climate change discourse on Twitter grew from a negligible 12 tweets in 2006 to a peak of 6,259,390 tweets in 2018 — growth of over 52 million percent across 14 years. Three distinct phases are visible in the data. The early phase from 2006 to 2012 saw rapid initial growth followed by decline between 2010 and 2012. The middle phase from 2013 to 2016 showed steady recovery coinciding with major climate policy milestones including the Paris Agreement of 2015. The recent phase from 2017 to 2019 produced the most explosive growth, peaking in 2018 at 6.26 million tweets — a single year representing 39.6% of all tweets in the entire 14-year dataset. The 2018 peak is strongly associated with the political fallout from the United States withdrawal from the Paris Agreement announced in 2017.

Year	Total Tweets	YoY Growth	Key Event
2006	12	—	Twitter launched — climate discourse begins
2007	3,961	+32,908%	IPCC Fourth Assessment Report published
2009	489,069	+1,531%	Copenhagen Climate Summit
2012	198,691	-37.49%	Discourse declines following Copenhagen disappointment

2015	1,259,359	+208.33%	Paris Agreement signed — global optimism peaks
2016	1,179,166	-6.37%	Post-Paris — sentiment remains high at +0.0458
2017	2,737,534	+132.16%	US announces withdrawal from Paris Agreement
2018	6,259,390	+128.65%	Peak year — largest climate Twitter discourse on record
2019	2,175,571	-65.24%	Discourse settles — sentiment turns negative at -0.0511

table 4: tweet volume by year with year-on-year (yoy) growth percentage and key contextual events

3.2 Stance Distribution

Across all 15,789,411 tweets, climate believers account for 71.52% of all climate discourse, neutrals for 20.94%, and deniers for just 7.55%. Believers have grown dramatically over 14 years from 16.67% in 2006 to 79.40% in 2018, while deniers declined from 8.33% to 6.18%. The 2015 Paris Agreement was the single most influential event in shifting public stance: believer percentage jumped by 23.27 percentage points in that year alone, the largest single-year shift in the entire dataset. Deniers, despite their minority status, generate aggression at nearly double the rate of believers and carry the most negative average sentiment in the dataset at -0.2320.

Stance	Total Tweets	Pct of Total	Avg Sentiment	Aggressive Pct
Believer	11,292,424	71.52%	+0.0168	27.33%
Neutral	3,305,601	20.94%	+0.0385	28.21%
Denier	1,191,386	7.55%	-0.2320	42.65%

table 5: stance distribution with average sentiment score and aggressive pct (percentage of tweets containing aggressive language)

3.3 Topic Distribution

Global Stance dominates climate Twitter discourse with 26.19% of all tweets, followed by Importance of Human Intervention at 16.43% and Weather Extremes at 15.61%. The topic landscape has shifted significantly over 14 years: Weather Extremes accounted for 36.01% of early-period tweets (2006-2012) but declined to just 9.77% in the recent period (2018-2019), as conversation evolved from discussing specific weather events to taking broader ideological positions. Politics grew from 5.76% to 12.80%, confirming that climate change became increasingly politicized over time. Significance of Pollution Awareness is the smallest topic at 3.04% but also the least aggressive at 15.95%, suggesting technical environmental discussions generate less hostility than political framing.

Topic	Total Tweets	Pct of Total	Avg Sentiment	Aggressive Pct
Global Stance	4,135,619	26.19%	+0.07	23.67%
Importance of Human Intervention	2,594,941	16.43%	+0.07	28.72%
Weather Extremes	2,464,814	15.61%	-0.08	24.32%
Politics	1,809,583	11.46%	-0.08	43.39%
Undefined / One Word Hashtags	1,305,118	8.27%	+0.24	25.67%
Donald Trump vs Science	996,244	6.31%	-0.16	40.30%
Seriousness of Gas Emissions	903,478	5.72%	-0.05	26.23%
Ideological Positions on Global Warming	602,695	3.82%	-0.20	34.88%
Impact of Resource Overconsumption	496,924	3.15%	-0.10	31.82%
Significance of Pollution Awareness	479,995	3.04%	-0.03	15.95%

table 6: topic distribution ranked by tweet volume — pct = percentage of all 15,789,411 tweets

3.4 Gender and Aggressiveness Distribution

Male authors account for 65.28% of all climate tweets, with female authors contributing 31.00% and undefined gender accounts making up 3.72%. The gender gap in aggressiveness is narrower than might be expected: males tweet aggressively 29.53% of the time compared to 27.38% for females. However, the most revealing gender finding is in the stance-aggressiveness intersection: female deniers are aggressive 42.39% of the time and male deniers 42.81%, confirming that denial-driven aggression is a stance-based phenomenon and not meaningfully influenced by gender. Overall, 71.33% of all tweets are classified as not aggressive with average sentiment of +0.0445, while the 28.67% of aggressive tweets carry substantially more negative sentiment at -0.1019.

4. Diagnostic Analytics

4.1 What Is Driving Aggressive Climate Discourse?

The most important diagnostic finding is that climate denial is the primary driver of aggressive and toxic discourse. Deniers tweet aggressively at 42.65%, nearly double the 27.33% rate of believers. Deniers also express the most negative average sentiment at -0.2320, compared to believers at +0.0168 and neutrals at +0.0385. Political topics amplify this dynamic: Politics (43.39% aggressive) and Donald Trump vs Science (40.30% aggressive) are the two most hostile topics in the dataset, both significantly above the platform average of 28.67%. In stark contrast, Significance of Pollution Awareness generates aggression at just 15.95%, confirming that technical environmental topics generate far less hostility than politically framed ones.

Segment	Aggressive Pct	Avg Sentiment	Key Insight
Politics topic	43.39%	-0.0848	Most aggressive topic in the entire dataset
Deniers (stance)	42.65%	-0.2320	Primary driver of toxic climate discourse
Trump vs Science topic	40.30%	-0.1597	Second most aggressive — politically charged
Ideological Positions topic	34.88%	-0.1977	Most negative avg sentiment of all topics
North America region	31.81%	-0.0219	Most aggressive continent with most deniers
Neutrals (stance)	28.21%	+0.0385	Slightly more aggressive than believers
Believers (stance)	27.33%	+0.0168	Least aggressive stance group
Asia region	17.93%	+0.1444	Least aggressive continent — most positive sentiment
Pollution Awareness topic	15.95%	-0.0291	Caldest topic — technical not political framing

table 7: aggressiveness analysis by segment — aggressive pct = percentage of tweets in that segment containing aggressive language

4.2 Sentiment Trends and the Paris Agreement Effect

Public sentiment followed a revealing trajectory across 14 years. From 2007 to 2014, average sentiment was consistently negative, oscillating between -0.049 and -0.017. A turning point occurred in 2015 when sentiment turned positive for the first time at +0.0248, rising to +0.0458 in 2016, the two years surrounding the Paris Agreement. This represents the strongest evidence in the dataset that the Paris Agreement generated genuine global optimism. However, this optimism was short-lived. Sentiment declined through 2017 and 2018 and turned sharply negative in 2019 at -0.0511, the most negative year in the high-volume period coinciding with growing evidence that Paris Agreement targets were not being met and increasing climate anxiety from accelerating extreme weather events. The implication is clear: policy announcements generate temporary sentiment uplift, but sustained positive sentiment requires visible implementation progress.

4.3 Regional Patterns and the North America Concentration

Regional analysis of the 5,307,538 geolocated tweets reveals a stark geographical divide. North America accounts for 65.1% of all geolocated tweets and is simultaneously the most aggressive region (31.81%), with the most negative average sentiment (-0.0219) and the highest concentration of deniers with 280,236 compared to just 7,221 in Asia and 47,495 in Europe. In stark contrast, Asia and Africa the regions most physically vulnerable to climate change maintain the most positive sentiment at +0.1444 and +0.1409 respectively, and the lowest aggressiveness rates. Europe's average

temperature deviation of +2.12 degrees Celsius is the highest of any region, suggesting European tweeters are experiencing the most pronounced warming effects relative to historical averages.

Continent	Geolocated Tweets	Avg Sentiment	Deniers	Aggressive Pct	Avg Temp Dev
North America	3,453,307	-0.0219	280,236	31.81%	+1.22°C
Europe	1,035,344	+0.0970	47,495	22.93%	+2.12°C
Oceania	333,090	+0.0074	23,053	29.48%	-0.23°C
Asia	263,876	+0.1444	7,221	17.93%	+0.73°C
Africa	141,016	+0.1409	3,949	22.44%	+0.51°C
South America	74,995	+0.0765	3,113	24.96%	N/A

table 8: regional analysis — geolocated tweets only (33.61% of full dataset) — temp dev = temperature deviation from historical average

4.4 Disaster Events and Tweet Volume Correlation

The disasters dataset contains 4,913 events affecting 203 countries with 730,807 recorded deaths and 2.38 billion people affected. Correlating disaster data with tweet sentiment reveals a counterintuitive finding: 2015 had the highest number of people affected by disasters at 428,839,661, yet sentiment was its most positive at +0.0248. This demonstrates that the Paris Agreement effect outweighed disaster-driven pessimism in that year. Conversely, 2019 saw 397 disaster events with 102,139,685 people affected, and sentiment was its most negative at -0.0511. As tweet volume grew dramatically from 2017 to 2018, aggressiveness actually decreased from 28.44% to 27.53% suggesting that a larger and more diverse Twitter audience diluted the aggressive minority's influence.

5. Recommendations

Recommendation 1 (Highest Priority) — Deploy Targeted Counter-Narrative Campaigns in North America Focused on Denier and Neutral Audiences

The data establishes beyond doubt that North America is the global epicenter of climate denial and aggressive discourse. With 280,236 identified deniers generating aggression at 42.65% nearly double the believer rate of 27.33% the region represents both the greatest risk to constructive climate dialogue and the greatest opportunity for audience conversion. The organization should commission a targeted social media communication strategy specifically designed for North American denier and neutral audiences. This strategy should avoid political framing entirely. Politics and Trump vs Science are the two most aggressive topics in the dataset and instead focus on Weather Extremes and Human Intervention, which generate substantially lower hostility at 24.32% and 28.72% respectively. Industry communication research consistently shows that values-based messaging framing climate action in terms of energy independence, economic opportunity, and community resilience rather than political ideology achieves significantly higher conversion rates among skeptical audiences. The Head of Growth should lead this initiative in partnership with a specialist North American communications

agency. Success should be measured by a reduction in the North American denier-to-believer ratio within 12 months, tracked using the same Twitter monitoring methodology applied in this analysis.

Evidence base: Table 7 (aggressiveness by segment), Table 8 (regional analysis), Section 4.1 and 4.3.

Recommendation 2 — Establish a Pre-Event Communication Protocol to Sustain Post-Agreement Sentiment Gains

The Paris Agreement effect documented in Section 4.2 demonstrates that major policy milestones generate measurable and sustained positive sentiment uplift, sentiment rose from -0.0094 in 2014 to +0.0458 in 2016, a shift of 0.055 sentiment units across 2.4 million tweets in two years. However, the subsequent decline to -0.0511 by 2019 reveals that announcement-driven optimism is fragile without visible implementation progress. The organization should develop a structured communication protocol to be activated six months before any major climate policy event COP summits, national climate legislation, or major scientific reports, and sustained for twelve months after. This protocol should include real-time social media monitoring to detect early sentiment shifts, proactive content publishing that translates policy commitments into concrete visible actions, and rapid response capabilities to address negative sentiment spikes before they compound. The Head of Policy Research should own this protocol with quarterly reviews tied to the sentiment metrics tracked in the Power BI dashboard delivered with this report.

Evidence base: Table 4 (tweet volume and key events), Section 4.2 (sentiment trends and Paris Agreement effect).

6. Data Quality Notes

6.1 Missing Geolocation Data (66.4% of Records)

The most significant data quality characteristic is that 10,481,873 tweets, 66.4% of all records lack coordinates. This is a structural feature of the Twitter platform: users must actively enable location sharing, and the majority choose not to. All regional and temperature-based analyses are therefore scoped to the 5,307,538 geolocated tweets (33.61% of total). If geolocated users are systematically different from the broader population for example, more urban or from specific demographics, then regional conclusions should be interpreted with appropriate caution. The has geolocation flag allows users to filter between the full dataset and the geolocated subset in the Power BI dashboard.

6.2 Dataset Coverage Gap: 2020-2022 Not Available

The dataset documentation describes coverage from 2008 to 2022. The version available for public download contains data from 2006 to 2019. The 2020-2022 period which includes the COVID-19 pandemic, the COP26 Glasgow summit of 2021, and accelerating extreme weather events cannot be assessed from this dataset. This is a known limitation of the Mendeley Data release and affects all analysts using this source. The 14-year period analyzed remains the most comprehensive longitudinal Twitter climate dataset available and provides robust trend analysis despite the coverage gap.

7. Conclusions

This analysis of 15,789,411 climate tweets across 14 years delivers a clear and actionable picture of global climate public opinion. Climate belief has won the majority of the public conversation, growing from 16.67% to 79.40%, but the discourse remains dangerously toxic, with 28.67% of all tweets containing aggressive language. The toxicity is concentrated in North America, driven disproportionately by the denier minority at 42.65% aggressiveness, and amplified by political framing of climate topics. The Paris Agreement demonstrated that major policy events can generate significant positive sentiment shifts, but the subsequent 2019 decline shows that implementation visibility is essential to sustaining those gains.

Two priority recommendations follow directly from these findings. The first is a targeted counter-narrative campaign in North America designed specifically for denier and neutral audiences, using non-political framing to reduce the hostility that currently dominates that region's climate discourse. The second is a structured communication protocol to maximize and sustain sentiment uplift around major climate policy milestones. Together, these two actions address the two most material gaps identified in the data regional discourse toxicity and sentiment fragility and are directly traceable to the evidence in this report. All underlying data, SQL queries, views, and interactive visualizations are accessible via the Power BI dashboard link provided separately.

8. Data Dictionary

Term	Definition
PCT	Percentage — used throughout this report as an abbreviation (e.g. aggressive pct = percentage of tweets in a segment containing aggressive language)
Sentiment Score	Numerical value from -1 (most negative) to +1 (most positive) indicating the emotional tone of a tweet. A score of 0 is perfectly neutral.
Avg Sentiment	Average (mean) sentiment score across all tweets in a given segment, year, topic, region, or stance group
Stance	The author's expressed position on climate change: Believer (accepts man-made climate change), Denier (rejects it), or Neutral (no clear position)
Aggressiveness	Binary label indicating whether a tweet contains aggressive or hostile language. Classified as Aggressive or Not Aggressive.
Aggressive Pct	The percentage of tweets within a given segment (topic, stance, region, gender) that are classified as Aggressive
Temperature Deviation	The difference in degrees Celsius between the average temperature at a tweet's location and the long-term historical average for that location. Positive values indicate warmer than average.
YoY Growth	Year-over-Year growth: the percentage change in a metric (e.g. tweet volume) between consecutive years
Geolocated Tweet	A tweet with valid longitude and latitude coordinates attached. Only 33.61% of tweets in this dataset are geolocated.

has_geolocation	A derived Boolean column added during data cleaning — TRUE if a tweet has valid coordinates, FALSE if coordinates are NULL
tweets_clean	The cleaned and transformed PostgreSQL table containing all 15,789,411 tweets with standardized values and derived columns
vw_yearly_trends	Aggregated PostgreSQL view summarizing tweet counts, sentiment, and stance metrics grouped by year — used in Power BI Page 1
vw_topic_analysis	Aggregated view summarizing tweet volume, sentiment, and aggressiveness grouped by topic — used in Power BI Page 2
vw_regional_analysis	Aggregated view summarizing discourse metrics grouped by continent for geolocated tweets only — used in Power BI Page 3
vw_tweets_vs_disasters	Joined view combining yearly tweet metrics with disaster event data for correlation analysis — used in Power BI Page 3
KPI	Key Performance Indicator — a quantifiable measure used to evaluate performance. KPI cards appear at the top of each dashboard page.
DOI	Digital Object Identifier — a permanent link to an academic paper. The dataset DOI is 10.1016/j.eswa.2022.117541

table 9: data dictionary — key terms and abbreviations used throughout this report

9. References

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